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• # Pin Number Definitions
• pwmPin = 18
• ledPin = 23
• butPin = 17
• # Function to simplify calling the sine function
• def sin(x):
•     return math.sin(x)
• # Linearly map an input scale to an output scale. This
can be used
• # to map the sin function (-1 to 1) into a duty cycle (0
to 100)%
• def mapFun(x, inLo, inHi, outLo, outHi):
•     inRange = inHi - inLo
•     outRange = outHi - outLo
•     inScale = (x - inLo) / inRange # normalised input (0
to 1)
•     return outLo + (inScale * outRange) # map normalised
input to output
•     GPIO.setmode(GPIO.BCM) # Use Broadcom pin-
numbering
•     GPIO.setup(ledPin,GPIO.OUT) # Setup an ouptut pins
•     GPIO.setup(pwmPin,GPIO.OUT)
•     # Setup input pin for button: use internal pull-up
resistor
•     GPIO.setup(butPin,GPIO.IN, pull_up_down=GPIO.PUD_UP)
•
•     pwm = GPIO.PWM(pwmPin, 200) # Initialise PWM channel
at 200Hz
•     GPIO.output(ledPin, GPIO.LOW) # LED off
•     pwm.start(0) # Initialise PWM with 0
duty
•     #### Main Loop ####
•     x = 0
•     try:
•         while 1:
•             # Decide how fast to breathe the LED
•             if GPIO.input(butPin): # Not
Pressed, breathe slow
•                 step = 0.03
•             else: # Pressed,
breathe fast

```